UC San Diego
JACOBS SCHOOL OF ENGINEERING
Shu Chien-Gene Lay Department of Bioengineering

From Circuits to Synapses

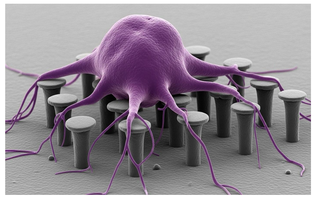
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What & Why?

Understanding synaptic integration requires resolving not only action potentials but also subthreshold membrane fluctuations. While patch-clamp methods provide intracellular access with high fidelity, they are low-throughput and invasive. Nanostructured electrode arrays (NEAs) promise scalable, minimally invasive recording, yet the signals they capture are often distorted and attenuated. These discrepancies arise from poorly understood interactions between the cell membrane, electrode interface, and amplifier.

Motivation: Bridging Recording Techniques



Patch Clamp → Nanostructured Arrays → MEA Recordings

Gold standard for signal fidelity

- Excellent SNR
- Limited to 1-2 cells at once
- Labor intensive

Promising Hybrid Approach

- Potential for intracellular-like signals
- Muti-site recording capability
- Signal fidelity issues that can be resolved

Extracellular approach

- Highly Scalable
  - Long-term recording
  - Limited signal resolution

Current Limitations

Microscopic Models Fall Short

- Traditional models assume uniform electrode-membrane contact
- Nanoscale electrodes create complex junctional regions
- Signal distortion remains poorly characterized
- Seal resistance dynamics not accurately captured

Designing better Nanoelectrodes through Circuit Characterization

Minimally Invasive

Nanoelectrodes interact with single cells without puncturing or rupturing cell membranes, preserving cellular integrity. Their morphology can also be tuned for specific cell types, such as neurons.

Circuit Modeling Approach

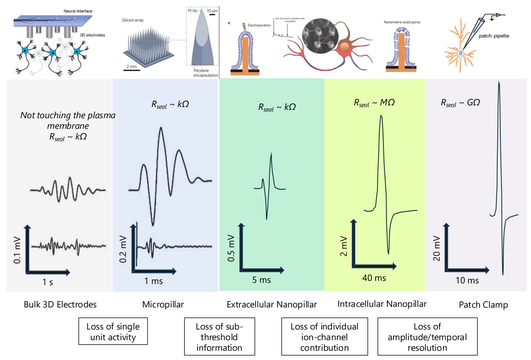
We adopt a circuit perspective, treating the cell-electrode interface as a transfer function. This enables accurate deconvolution of recorded signals based on the complex cell-electrode coupling in real-time.

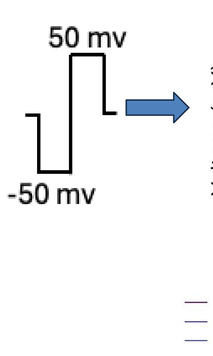
Objective

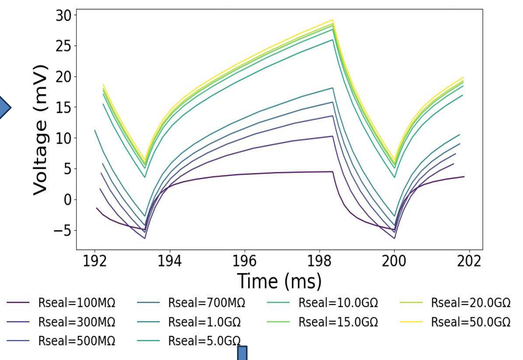


We develop new circuit-based models to uncover how these coupled elements shape recorded neural waveforms.

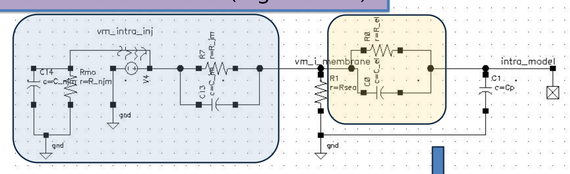
But… what did we actually do?



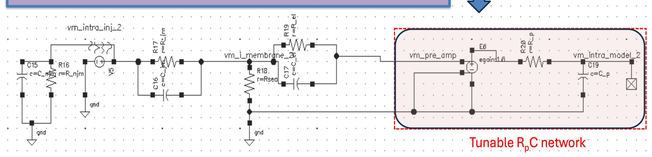


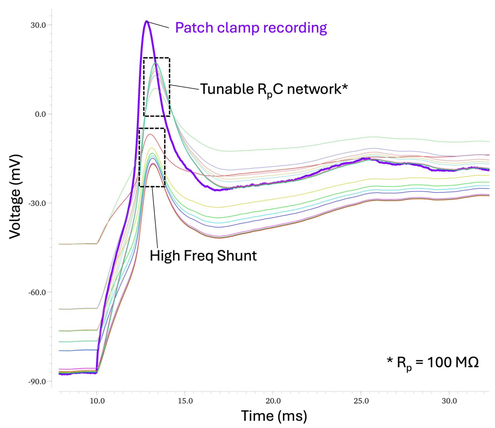


Base Intracellular Model (High-f shunt)



Enhanced Intracellular Model (RpC Network)





1. Attenuation Reduced

The RpC network minimizes signal loss, preserving crucial amplitude and waveform characteristics that were previously distorted by simplistic models.

2. High-Frequency Shunt Corrected

Accurate representation of parasitic elements mitigates the high-frequency shunting effect, allowing for more accurate capture of rapid action potential dynamics.

3. Closer Match to Patch Clamp

The modified model's output demonstrates better agreement with gold-standard patch-clamp recordings, validating its ability to represent true intracellular signals.

Amplitude Attenuation

- High Rseal (>5 GΩ) preserves AP amplitude
- Low Rseal (<500 MΩ) reduces signal significantly

Temporal Broadening

- AP half-width increases by 1.3-2.8x depending on electrode parameters
- Fast components (rise time) are disproportionally affected

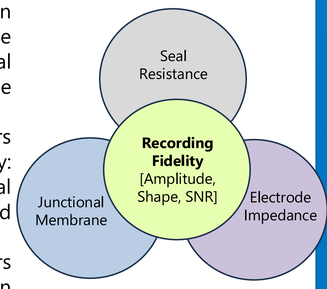
Filtering Effects

- Electrode capacitance creates a high shunt
- Critical frequency cutoff scales with Rseal

What does it mean though?

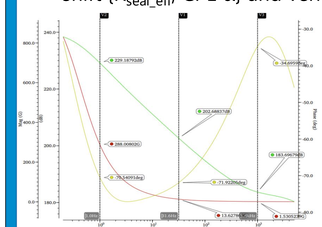
Discussion

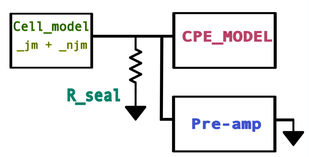
- Circuit-based modeling can provide deep quantitative understanding of signal transformation at the nanoscale
- Three critical factors determine recording quality: seal resistance, junctional membrane properties, and electrode impedance
- Optimizing these parameters can bridge the gap between patch-clamp fidelity and nanoelectrode scalability



Future Direction

- Integrate a **CPE** interface model based on an R-C ladder
- Replace R_{seal} with a **finite R-C ladder** to approximate the cleft's disturbed impedance
- Predict how coatings/geometry (tip radius, height, porosity) shift {R_{seal_eff}, CPE α} and verify with new EIS + physiology.





Simulation Methodology

Parameter Selection

- Membrane cap: 1μF/cm²
- R_{jm} (junctional membrane): 100MΩ
- R_{njm} (non-junct. Membrane): 50M – 5GΩ
- Seal Resistance: 100MΩ – 10GΩ
- Electrode Cap: 8 pF

Simulation Tools

- Circuit simulation in Cadence
- Python for parameter sensitivity testing and waveform analysis

Validation Method:

- Comparison to experimental recordings from cultured neurons
- Statistical fitting of model parameters in cadence
- Monte-carlo analysis of parameter uncertainty

& Acknowledgement

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INTEGRATED SYSTEMS NEUROENGINEERING

Jahed Bio-Nano-Electronics Lab